

Notice that we cannot claim an equational proof in the case of reduced rings, since they do not form an algebraic theory.

Proof. Let $e^2 = e$ and x be any element. We compute

$$(exe - ex)^2 = exeexe - exeex - exexe + exex = exeexe - exeex - exexe + exex = 0.$$

Since the rng is reduced, we get $exe = ex$. A similar calculation shows $exe = xe$. Hence, e commutes with x . In an n -rng, we can make this argument equational: From $(exe - ex)^2 = 0$ we conclude $exe - ex = (exe - ex)^n = 0$ (since $n > 1$), and proceed exactly as before. $\square$

Lemma 2.9. *Let A be a rng, $n > 1$ and $x \in A$ be an element with $x^n = x$. If $k, k' \geq 1$ satisfy $k \equiv k' \pmod{n-1}$, then $x^k = x^{k'}$ by an equational proof. In particular, x^{n-1} is idempotent by an equational proof.*

Proof. Based on $x^{n-1}x = x$, a simple induction shows $x^{(n-1)i+j} = x^j$ for all $i \geq 0, j \geq 1$. $\square$

Corollary 2.10. *If $n, m > 1$ are such that $m-1 \mid n-1$, then every m -ring is an n -ring by an equational proof.* $\square$

The last three results will be used a lot in the following. Let us give two sample applications:

Example 2.11. Here is an equational proof that every 3-rng is commutative, taken from [BM13]. Let x be any element. By Lemmas 2.8 and 2.9 every square y^2 is central. In particular, $(x^2 + x)^2$ is central. It simplifies to $x^4 + 2x^3 + x^2 = x^2 + 2x + x^2 = 2x^2 + 2x$. Since x^2 is central, it follows that $2x$ is central. Applying $(x^2 + x)^2 = 2(x^2 + x)$ twice, we get $x^2 + x = (x^2 + x)^3 = 4(x^2 + x)$, hence $3x^2 + 3x = 0$. Since x^2 is central, it follows that $3x$ is central. Thus, $x = 3x - 2x$ is central as well.

Example 2.12. The following equational proof that every 4-rng is commutative is inspired from [Wa99]. Let x be any element. Then $-x = x$, since 4 is even. Also, every cube x^3 is central by Lemmas 2.8 and 2.9. We claim that $f(x) := x + x^2$ is idempotent, and hence central. In fact, it squares to $x^2 + 2x^3 + x^4 = x^2 + 0 + x = x + x^2$. For all elements x, y then also $f(x+y) - f(x) - f(y)$ is central, which simplifies to $xy + yx$. In particular, $x(xy + yx) = (xy + yx)x$, which simplifies to $x^2y = yx^2$. So y commutes with x^2 . Since $x + x^2$ is central, y also commutes with x . (We will generalize this method in Theorem 6.4.)

Remark 2.13. As explained in the introduction, every equational proof of the commutativity of n -rings leads to a representation of the commutator $XY - YX$ as a sum of expressions of the form $g \cdot (f^n - f) \cdot h$ in the free ring $\mathbb{Z}\langle X, Y \rangle$. By unwinding what happens in Example 2.11, we get the following equation, where $p(f) := f^3 - f$.

$$\begin{aligned} XY - YX = & \\ & + (p(X + X^2) - p(X)(4 + 4X + 3X^2 + X^3))Y - Y(p(X + X^2) - p(X)(4 + 4X + 3X^2 + X^3)) \\ & + 3p(X^2YX^2 - YX^2) - 3p(X^2YX^2 - X^2Y) \\ & + 3(X^2YX^2 - X^2Y)X^2YXp(X)Y(X^2 - 1) - 3(X^2YX^2 - YX^2)(X^2 - 1)YXp(X)YX^2 \\ & + p((X + X^2)^2Y(X + X^2)^2 - Y(X + X^2)^2) - p((X + X^2)^2Y(X + X^2)^2 - (X + X^2)^2Y) \\ & + ((X + X^2)^2Y(X + X^2)^2 - (X + X^2)^2Y)(X + X^2)^2Y(X + X^2)p(X + X^2)Y((X + X^2)^2 - 1) \\ & - ((X + X^2)^2Y(X + X^2)^2 - Y(X + X^2)^2)((X + X^2)^2 - 1)Y(X + X^2)p(X + X^2)Y(X + X^2)^2 \\ & + 2p(X^2YX^2 - X^2Y) - 2p(X^2YX^2 - YX^2) \\ & + 2(X^2YX^2 - YX^2)(X^2 - 1)YXp(X)YX^2 - 2(X^2YX^2 - X^2Y)X^2YXp(X)Y(X^2 - 1) \\ & + (2 + X)p(X)Y - Y(2 + X)p(X) \end{aligned}$$

It shows that every 3-ring is commutative. Even though this is a very bad way of writing down the proof, it is interesting that such a single equation always exists. And it also shows for $d \in \mathbb{Z}$ that a ring satisfying $dx = dx^3$ for all x satisfies $dxy = dyx$ for all x, y .

Proposition 2.14. *Commutativity can be reduced to the unital case. More precisely:*

- (1) *If every potent ring is commutative, then every potent rng is commutative.*
- (2) *If every n -ring is commutative by an equational proof, then every n -rng is commutative by an equational proof.*

Proof. 1. Let A be a potent rng. Let $x, y \in A$ and choose $n, m > 1$ with $x^n = x$ and $y^m = y$. By Lemmas 2.8 and 2.9 the elements x^{n-1} and y^{m-1} are central. Consider the element

$$e := x^{n-1} + y^{m-1} - x^{n-1}y^{m-1} = y^{n-1} + x^{m-1} - y^{n-1}x^{m-1}.$$

We calculate

$$ex = x^{n-1}x + y^{m-1}x - x^{n-1}y^{m-1}x = x^n + y^{m-1}x - x^n y^{m-1} = x + y^{m-1}x - xy^{m-1} = x.$$

A similar calculation shows $ey = y$. By symmetry we also get $xe = x$ and $ye = y$. Hence, if B denotes the subrng of A generated by x, y , which consists of $\mathbb{Z}$ -linear combinations of finite non-empty products of x and y , then e is multiplicative identity of B , and clearly B is potent. By assumption, B is commutative, so that $xy = yx$.

2. In the case of n -rings, we can take $m = n$ and define e as above. There is an equational proof of $xy = yx$ in the theory of n -rings. We may assume that x, y are the only variables in the proof. Since e is neutral for x and y , every appearance of the multiplicative identity can be replaced by e . This yields an equational proof of $xy = yx$ in the theory of n -rngs. $\square$

Let us demonstrate this reduction in another special case:

Lemma 2.15. *In an n -rng, there is an equational proof of*

$$\sum_{k=1}^{n-1} \binom{n}{k} x^k = 0.$$

Proof. If we had a multiplicative identity, then this follows simply by expanding $1+x = (1+x)^n$ with the binomial theorem and then cancelling 1 and $x^n = x$. If not, we just do the same with $x^{n-1} + x = (x^{n-1} + x)^n$ and use that x^{n-1} is neutral for all powers of x . $\square$

Because of Proposition 2.14 we will only work with rings (i.e. unital rings) from now on.

Remark 2.16. The following are equivalent for an element x of an n -ring.

- (1) x is a unit.
- (2) x is no zero-divisor.
- (3) $x^{n-1} = 1$.

Lemma 2.17. *Let A be an n -ring. Then every element $x \in A$ can be written as $x = e \cdot u$, where $e \in A$ is idempotent and $u \in A$ is a unit. Namely, we have $e = x^{n-1}$ and $u = x + (1 - x^{n-1})$.*

Proof. The element $e := x^{n-1}$ is idempotent by Lemma 2.9. Let $u := x + (1 - e)$. Then we have $eu = ex = x^n = x$. We need to prove that u is a unit. We use Remark 2.16. Notice that the product of x and $1 - e$ is zero, and $1 - e$ is idempotent. Hence, the binomial theorem shows $u^{n-1} = x^{n-1} + (1 - e)^{n-1} = x^{n-1} + (1 - e) = 1$. Alternatively, one may verify directly that $x^{n-2} + (1 - e)$ is inverse to u . $\square$

Corollary 2.18. *Let A be an n -ring. If every unit of A commutes with every unit of A (by an equational proof), then A is commutative (by an equational proof).*

Proof. This follows from Lemmas 2.8 and 2.17. $\square$

n	n -fields	n	n -fields	n	n -fields
2	$\mathbb{F}_2$	18	$\mathbb{F}_2$	34	$\mathbb{F}_2, \mathbb{F}_4$
3	$\mathbb{F}_2, \mathbb{F}_3$	19	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4, \mathbb{F}_7, \mathbb{F}_{19}$	35	$\mathbb{F}_2, \mathbb{F}_3$
4	$\mathbb{F}_2, \mathbb{F}_4$	20	$\mathbb{F}_2$	36	$\mathbb{F}_2, \mathbb{F}_8$
5	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5$	21	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_{11}$	37	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4, \mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_{13}, \mathbb{F}_{19}, \mathbb{F}_{37}$
6	$\mathbb{F}_2$	22	$\mathbb{F}_2, \mathbb{F}_4, \mathbb{F}_8$	38	$\mathbb{F}_2$
7	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4, \mathbb{F}_7$	23	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_{23}$	39	$\mathbb{F}_2, \mathbb{F}_3$
8	$\mathbb{F}_2, \mathbb{F}_8$	24	$\mathbb{F}_2$	40	$\mathbb{F}_2, \mathbb{F}_4$
9	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_9$	25	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4, \mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_9, \mathbb{F}_{13}, \mathbb{F}_{25}$	41	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_9, \mathbb{F}_{11}, \mathbb{F}_{41}$
10	$\mathbb{F}_2, \mathbb{F}_4$	26	$\mathbb{F}_2$	42	$\mathbb{F}_2$
11	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_{11}$	27	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_{27}$	43	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4, \mathbb{F}_7, \mathbb{F}_8, \mathbb{F}_{43}$
12	$\mathbb{F}_2$	28	$\mathbb{F}_2, \mathbb{F}_4$	44	$\mathbb{F}_2$
13	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4, \mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_{13}$	29	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_8, \mathbb{F}_{29}$	45	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_{23}$
14	$\mathbb{F}_2$	30	$\mathbb{F}_2$	46	$\mathbb{F}_2, \mathbb{F}_4, \mathbb{F}_{16}$
15	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_8$	31	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4, \mathbb{F}_7, \mathbb{F}_{11}, \mathbb{F}_{16}, \mathbb{F}_{31}$	47	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_{47}$
16	$\mathbb{F}_2, \mathbb{F}_4, \mathbb{F}_{16}$	32	$\mathbb{F}_2, \mathbb{F}_{32}$	48	$\mathbb{F}_2$
17	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_9, \mathbb{F}_{17}$	33	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_9, \mathbb{F}_{17}$	49	$\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4, \mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_9, \mathbb{F}_{13}, \mathbb{F}_{17}, \mathbb{F}_{25}, \mathbb{F}_{49}$

TABLE 2. List of n -fields for $2 \leq n \leq 49$

3. Structure of n -rings

In this section we will investigate the well-known structure of (commutative) n -rings. We also find an equational proof of the reduction to prime characteristic.

Remark 3.1. The n -fields are those $\mathbb{F}_q$, where q is a prime power satisfying $q - 1 \mid n - 1$. This is because $\mathbb{F}_q^\times$ is cyclic of order $q - 1$. See Table 2 for a list of examples. It has been generated programmatically with a bit of SageMath¹ code, see appendix A.1. It follows in particular that every n -ring of characteristic p satisfies $p - 1 \mid n - 1$ (since it contains $\mathbb{F}_p$ which is then also an n -ring).

It turns out that n -fields generate all commutative n -rings in a suitable sense:

Proposition 3.2. *A commutative ring is an n -ring if and only if it is a subring of a direct product of fields of the form $\mathbb{F}_q$, where $q - 1 \mid n - 1$.*

Proof. The direction $\Leftarrow$ is clear. To prove $\Rightarrow$, let A be a commutative n -ring. Then A is reduced, which means that the intersection of all prime ideals is zero. Hence, the canonical map $A \rightarrow \prod_{\mathfrak{p}} Q(A/\mathfrak{p})$ is injective, where $\mathfrak{p}$ runs through all prime ideals of A . Since $Q(A/\mathfrak{p})$ (which is just $A/\mathfrak{p}$, since $\dim(A) \leq 0$) is clearly an n -field, the claim follows. $\square$

Practically, this means that the class of commutative n -rings is already determined by the class of n -fields. For example, we have the following improvement of Corollary 2.10:

Corollary 3.3. *Let $n, n' > 1$. Every commutative n -ring is a commutative n' -ring if and only if $q - 1 \mid n - 1 \implies q - 1 \mid n' - 1$ holds for all prime powers q .*

Proof. The direction $\Leftarrow$ follows from Proposition 3.2. The direction $\Rightarrow$ follows by considering the finite field $\mathbb{F}_q$. $\square$

Example 3.4. Every 10-ring is a 4-ring, and every 14-ring is a 2-ring; see also Table 2.

¹<https://www.sagemath.org/>